03-PRODUCT-REQUIREMENTS-DOCUMENT.md

.things that Matter

Product Requirements Document (PRD)

Document Information	
Company Name	.things that Matter
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1. Product Vision

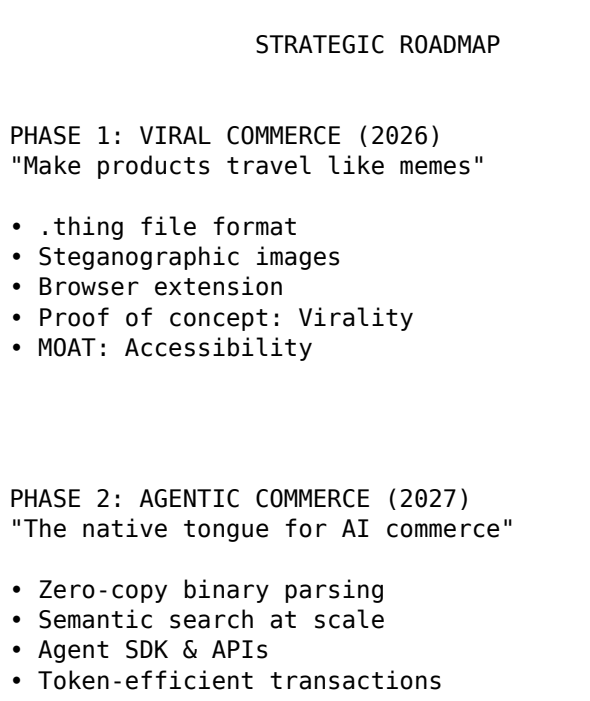
1.1 Vision Statement

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1.2 Strategic Phases

Phase	Codename	Focus	Moat
Phase 1	**Viral Commerce**	Create .thing files and establish virality	**Accessibility**
Phase 2	**Agentic Commerce**	Optimize for AI efficiency	**Energy Efficiency**
Phase 3	**Decentralized Commerce**	Blockchain-based open infrastructure	**Decentralization**

1.3 Phase Overview



• MOAT: Energy Efficiency

PHASE 3: DECENTRALIZED COMMERCE (2028+)
"Open protocol for the world"

• Blockchain-based registry
• Smart contract checkout
• Open-source everything
• Community governance
• MOAT: Decentralization

2. Phase 1: Viral Commerce

2.1 Phase 1 Overview

Attribute	Value
Timeline	2026 (11 months)
Budget	R72,000
Goal	Prove that products can travel virally across the internet
Success Metric	10,000 .thing files created, 100+ extension users
Moat	Accessibility — products available everywhere

2.2 Phase 1 Problem Statement

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- Products are trapped in platform silos
- Sharing a product = sharing a link to a storefront
- AI agents waste compute scraping HTML to understand products
- No universal format for portable commerce

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- Products exist as self-contained digital objects
- Sharing a product = sharing the product itself (in an image, link, or file)
- Anyone can decode a product anywhere
- Commerce becomes a layer of reality, not a destination

2.3 Phase 1 Requirements

2.3.1 Functional Requirements

ID	Requirement	Priority	Description
P1-F01	.thing File Format	Must Have	Binary format using FlatBuffers schema
P1-F02	Commerce Compiler	Must Have	Transform product JSON → .thing files
P1-F03	CLIP Vectorization	Must Have	Generate 512-dim semantic embeddings
P1-F04	Ed25519 Signing	Must Have	Cryptographic tamper-proof seal
P1-F05	Zstd Compression	Must Have	Compress descriptions for portability
P1-F06	Polyglot Images	Must Have	Embed .thing in images (steganography)
P1-F07	XMP Metadata Beacon	Must Have	Fast discovery layer in image headers

P1-F08	matter:// URI	Must Have	Text-based distribution protocol
P1-F09	Shadow Index API	Must Have	Hash → artifact resolution service
P1-F10	Browser Extension	Must Have	Detect and display Matter products
P1-F11	Cloudflare R2 CDN	Must Have	Artifact storage and distribution
P1-F12	Batch Compilation	Should Have	Process 1000+ products at once
P1-F13	Merchant Dashboard	Could Have	Self-service product compilation
P1-F14	Analytics	Could Have	Track .thing views and clicks

2.3.2 Non-Functional Requirements

ID	Requirement	Target	Description
P1-NF01	Compile Latency	<500ms	Time to compile single product
P1-NF02	File Size	<50KB	Average .thing file size
P1-NF03	Extension Load	<100ms	Time to detect Matter content
P1-NF04	Signature Verify	<10ms	Client-side verification time
P1-NF05	CDN Availability	99.9%	Artifact availability SLA
P1-NF06	Steg Survival	≥50%	Platforms where payload survives

2.3.3 Phase 1 User Stories

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ID	As a...	I want to...	So that...
P1-US01	Merchant	Upload my product catalog	My products become portable .thing files
P1-US02	Merchant	Download polyglot images	I can share shoppable images on social media
P1-US03	Merchant	Get matter:// links	I can share products in text-based channels
P1-US04	Merchant	See which products are compiled	I know my catalog is ready
P1-US05	Merchant	Update product prices	The .thing files reflect current pricing

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ID	As a...	I want to...	So that...
P1-US06	Consumer	Install a browser extension	I can discover Matter products anywhere
P1-US07	Consumer	See product info on any image	I don't need to visit a store
P1-US08	Consumer	Click to buy directly	I can purchase without navigation
P1-US09	Consumer	Trust the price shown	I know it hasn't been tampered with
P1-US10	Consumer	Share a product with friends	They can also see the product info

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ID	As a...	I want to...	So that...
P1-US11	Developer	Access a compilation API	I can integrate Matter into my app
P1-US12	Developer	Read .thing files easily	I can build Matter-compatible tools
P1-US13	Developer	Verify signatures	I can trust the data integrity

2.4 Phase 1 Acceptance Criteria

Milestone	Acceptance Criteria
Phase 0 Complete	Steganography tested on 20 platforms, ≥50% survival rate confirmed
Compiler MVP	Successfully compile JSON → .thing → polyglot image
CLIP Working	Semantic vectors embedded in .thing files
Extension Alpha	Detect XMP beacon, display product overlay
75% Complete	End-to-end demo: image → decode → product display
Beta Launch	5 merchants onboarded, 1000+ products compiled
Public Launch	Extension in Chrome Web Store, documentation complete

3. Phase 2: Agentic Commerce

3.1 Phase 2 Overview

Attribute	Value
Timeline	2027
Budget	TBD (based on Phase 1 revenue)
Goal	Make .thing the most efficient format for AI agent commerce
Success Metric	100K+ agent transactions, documented efficiency gains
Moat	Energy Efficiency — cheaper for AI to use

3.2 Phase 2 Problem Statement

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- AI agents scrape HTML pages to understand products (expensive)
- JSON APIs are verbose and require full parsing (inefficient)
- Each platform has different data structures (fragmented)
- Agents waste tokens on data extraction (costly)

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- .thing is the "native tongue" for AI commerce
- Zero-copy field access means O(1) reads (fast)
- Semantic vectors enable instant similarity search (smart)
- Standardized format across all merchants (unified)
- Agents use fewer tokens and less compute (cheap)

3.3 Phase 2 Requirements

3.3.1 Functional Requirements

ID	Requirement	Priority	Description
P2-F01	Agent SDK	Must Have	Rust/JS libraries optimized for AI agents
P2-F02	Bulk Read API	Must Have	Efficient batch fetching of .thing files
P2-F03	Vector Search API	Must Have	Query products by semantic similarity
P2-F04	Streaming Compilation	Must Have	Real-time catalog sync for merchants
P2-F05	WASM Runtime	Must Have	Run compiler in any environment
P2-F06	Token Estimation	Should Have	Help agents budget API calls
P2-F07	Agent Analytics	Should Have	Track agent usage patterns
P2-F08	Rate Limiting	Should Have	Fair usage across agents
P2-F09	Caching Layer	Should Have	Reduce redundant fetches
P2-F10	GraphQL API	Could Have	Flexible querying for agents

3.3.2 Non-Functional Requirements

ID	Requirement	Target	Description
P2-NF01	Read Latency	<5ms	Time to read single field from .thing
P2-NF02	Vector Search	<50ms	Similarity search across 100K products
P2-NF03	Throughput	10K req/s	API requests per second
P2-NF04	Token Efficiency	90% reduction	vs. HTML scraping
P2-NF05	Compute Cost	80% reduction	vs. JSON parsing

3.3.3 Phase 2 User Stories

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ID	As an...	I want to...	So that...
P2-US01	AI Agent	Read product prices in O(1) time	I minimize compute costs
P2-US02	AI Agent	Search products by semantic meaning	I find relevant items without key words
P2-US03	AI Agent	Verify product authenticity	I don't transact with fake listings
P2-US04	AI Agent	Batch fetch product catalogs	I efficiently sync data
P2-US05	AI Agent	Estimate token usage before calls	I budget my operations

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ID	As a...	I want to...	So that...
P2-US06	AI Platform	Integrate Matter SDK	My agents can do commerce efficiently
P2-US07	AI Platform	Monitor agent transactions	I understand usage patterns
P2-US08	AI Platform	Set rate limits	I ensure fair resource usage

3.4 Phase 2 Efficiency Metrics

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Operation	Traditional (HTML/JSON)	Matter (.thing)	Improvement
Parse product page	~500ms, O(n)	~5ms, O(1)	**100x faster**
Read price field	Parse entire response	Direct byte offset	**Zero allocation**
Semantic search	External API call	Local vector math	**No API cost**
Verify authenticity	Trust platform	Cryptographic proof	**Trustless**
Token consumption	~2000 tokens/product	~50 tokens/product	**40x cheaper**
Memory usage	Full DOM/JSON	Memory-mapped file	**90% reduction**

4. Phase 3: Decentralized Commerce

4.1 Phase 3 Overview

Attribute	Value
Timeline	2028+
Budget	TBD (community-funded / token)
Goal	Open-source the protocol and decentralize the infrastructure
Success Metric	Protocol adoption by 1000+ independent nodes
Moat	Decentralization — no single point of control

4.2 Phase 3 Problem Statement

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- .things that Matter controls the signing keys
- Shadow Index is a centralized service
- CDN is our infrastructure
- Users must trust our company

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- Anyone can become a signing authority
- Product registry lives on blockchain
- Artifacts stored on IPFS/Filecoin

- Protocol governed by community
- Trust is cryptographic, not institutional

4.3 Phase 3 Requirements

4.3.1 Functional Requirements

ID	Requirement	Priority	Description
P3-F01	Open Source Release	Must Have	All code MIT/Apache licensed
P3-F02	Decentralized Registry	Must Have	Blockchain-based product index
P3-F03	Multi-Signer Support	Must Have	Multiple trusted signing authorities
P3-F04	IPFS Integration	Must Have	Decentralized artifact storage
P3-F05	Smart Contract Checkout	Must Have	On-chain payment execution
P3-F06	Governance Token	Should Have	Community decision-making
P3-F07	Node Software	Should Have	Anyone can run a Matter node
P3-F08	Protocol Specification	Must Have	Formal spec for interoperability
P3-F09	Validator Network	Could Have	Decentralized compilation validation
P3-F10	Cross-Chain Bridge	Could Have	Multi-blockchain support

4.3.2 Non-Functional Requirements

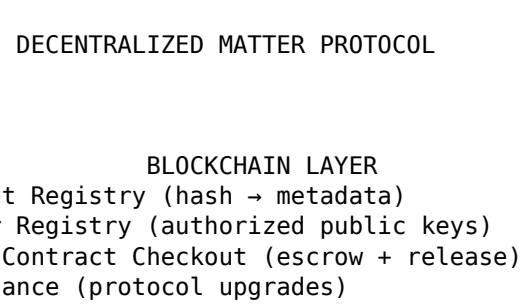
ID	Requirement	Target	Description
P3-NF01	Decentralization	No single point of failure	Full redundancy
P3-NF02	Censorship Resistance	Cannot be taken down	Distributed infrastructure
P3-NF03	Permissionless	Anyone can participate	Open protocol
P3-NF04	Backwards Compatible	Existing .thing files work	Smooth migration

4.3.3 Phase 3 User Stories

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ID	As a...	I want to...	So that...
P3-US01	Developer	Fork and modify the protocol	I can innovate on commerce infrastructu
re
P3-US02	Node Operator	Run a Matter node	I earn fees and support the network
P3-US03	Merchant	Choose my signing authority	I'm not dependent on one company
P3-US04	Token Holder	Vote on protocol changes	I have a say in governance
P3-US05	Consumer	Trust math, not companies	My transactions are verifiable

4.4 Phase 3 Architecture Vision



- IPFS (artifact storage)
- Filecoin (persistence incentives)
- Arweave (permanent storage option)

NODE NETWORK

- Compiler Nodes (run matter-core)
- Index Nodes (serve Shadow Index)
- Gateway Nodes (HTTP access for legacy)

CLIENT LAYER

- Browser Extension (unchanged UX)
- Agent SDKs (same APIs)
- Mobile Apps (native clients)

5. Feature Specifications (Phase 1 Detail)

5.1 .thing File Format

\1 Universal Commerce Container for portable product data.

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Field	Type	Required	Description
magic	uint32	Yes	0x54484E47 ("THNG")
version	uint8	Yes	Schema version (1)
id	string	Yes	Unique product identifier
hash	bytes	Yes	SHA-256 of canonical form
title	string	Yes	Product name
description	string	No	Product description (Zstd compressed)
price	uint64	Yes	Price in cents
currency	string	Yes	ISO 4217 currency code
sku	string	No	Merchant SKU
image_url	string	No	Primary image URL
clip_vector	float[512]	Yes	Semantic embedding
checkout_url	string	Yes	Purchase destination
merchant_id	string	No	Merchant identifier
created_at	uint64	Yes	Unix timestamp
expires_at	uint64	No	TTL for freshness
signature	bytes	Yes	Ed25519 signature
signer_pubkey	bytes	Yes	Signer's public key

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Content	Size
Header + metadata	~500 bytes
CLIP vector	~2KB
Compressed description	~1-5KB
Signature	64 bytes
Total typical	**3-8KB**

5.2 Commerce Compiler

\1 Transform product data into .thing files and distribution assets.

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```
{
  "id": "prod_123",
  "title": "Premium Wireless Headphones",
  "description": "High-fidelity audio with 30-hour battery...",
  "price": 14999,
  "currency": "USD",
  "image_url": "https://example.com/headphones.jpg",
  "checkout_url": "https://store.example.com/checkout/prod_123"
}
```

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```
{
  "hash": "QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "thing_url": "https://r2.thingsthatmatter.com/QmYwAP...thing",
  "image_url": "https://r2.thingsthatmatter.com/QmYwAP...jpg",
  "matter_uri": "matter://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "gist": {
    "title": "Premium Wireless Headphones",
    "price": 14999,
    "currency": "USD"
  }
}
```

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1. \1 — Check input against schema
2. \1 — Download product image
3. \1 — Send to CLIP API, receive 512-dim vector
4. \1 — Zstd compress description
5. \1 — Build FlatBuffers binary
6. \1 — SHA-256 of canonical binary
7. \1 — Ed25519 signature
8. \1 — Steganographic image creation
9. \1 — Push to Cloudflare R2
10. \1 — Record in Shadow Index

5.3 Browser Extension (Lens)

\1 Detect and display Matter products on any webpage.

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Feature	Description
XMP Detection	Scan images for Matter metadata beacons
URI Interception	Handle matter:// protocol links
Overlay Display	Show product card on hover/click
Signature Verification	Validate Ed25519 in WASM
Local Cache	IndexedDB storage for resolved products
Settings	Enable/disable, visual preferences

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```
  [Product Image]

  Premium Wireless Headphones
  $149.99 USD

  [ Verified by .things that Matter]


           BUY NOW


  Powered by Matter Protocol
```

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Permission	Reason
activeTab	Read page content for detection
storage	Cache resolved products
webRequest	Intercept matter:// URIs

5.4 Shadow Index API

\1 Resolve product hashes to artifact locations.

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Method	Endpoint	Description
POST	/v1/compile	Compile product JSON to .thing
POST	/v1/batch	Compile multiple products
GET	/v1/resolve/{hash}	Get artifact URLs for hash
GET	/v1/gist/{hash}	Get minimal product data
GET	/v1/health	Service health check

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```
GET /v1/resolve/QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG

Response:
{
  "hash": "QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG",
  "thing_url": "https://r2.thingsthatmatter.com/QmYwAP.thing",
  "image_url": "https://r2.thingsthatmatter.com/QmYwAP.jpg",
  "gist": {
    "title": "Premium Wireless Headphones",
    "price": 14999,
    "currency": "USD"
  },
  "created_at": 1706000000,
  "expires_at": 1737600000
}
```

6. Success Metrics

6.1 Phase 1 KPIs

Metric	Target	Measurement
.thing files created	10,000	Database count
Unique products compiled	5,000	Distinct SKUs
Extension installs	100+	Chrome Web Store
Active extension users	50+	Weekly active
Merchants onboarded	5-10	Signed agreements
Steganography survival	≥50%	Platform testing
Compile success rate	>99%	API metrics
Average compile time	<500ms	API metrics
Extension detection rate	>95%	Test suite

6.2 Phase 2 KPIs (Future)

Metric	Target	Measurement
Agent SDK downloads	1,000+	Package registry
Agent transactions	100K+	API logs
Token efficiency gain	40x	Benchmark tests
Compute cost reduction	80%	Benchmark tests
Vector search queries	1M+	API metrics

6.3 Phase 3 KPIs (Future)

Metric	Target	Measurement
Independent nodes	1,000+	Network discovery
Open source contributors	100+	GitHub metrics
Protocol forks	50+	GitHub forks
Decentralized transactions	1M+	Blockchain data

7. Dependencies & Integrations

7.1 External Dependencies

Dependency	Purpose	Phase	Risk
Replicate API	CLIP inference	P1	Medium
Cloudflare R2	Artifact storage	P1	Low
Supabase	Database	P1	Low
Vercel	API hosting	P1	Low
Chrome Web Store	Extension distribution	P1	Low
IPFS	Decentralized storage	P3	Medium
Ethereum/Polygon	Smart contracts	P3	Medium

7.2 Integration Points

Integration	Type	Phase
Merchant product catalogs	Inbound data	P1
Social media platforms	Image distribution	P1
AI agent frameworks	SDK integration	P2
Blockchain networks	Protocol layer	P3

8. Constraints & Assumptions

8.1 Constraints

Constraint	Impact
R72,000 Year 1 budget	Lean development, free tiers
Solo developer	Sequential work
11-month Phase 1 timeline	Extended milestones
No security audit budget	Self-review initially

8.2 Assumptions

Assumption	Validation
Steganography survives ≥50% of platforms	Phase 0 testing
CLIP API remains affordable	Monitor pricing
Chrome allows extension	Review policies
Merchants want portable products	Beta feedback
AI agents will adopt efficient formats	Market trends

9. Glossary

Term	Definition
.thing file	Universal Commerce Container binary format
Matter Protocol	The overall system for portable commerce
Polyglot Image	Image file that is also a valid .thing file
Shadow Index	Hash-to-artifact resolution service
Lens	Browser extension for Matter detection
CLIP Vector	512-dimension semantic embedding
Zero-Copy Access	Reading data without memory allocation
Steganography	Hiding data within images
XMP Beacon	Metadata header for fast discovery
matter:// URI	Custom protocol for text-based sharing

10. Document Approval

Role	Name	Signature	Date
CEO	Sobuhle Nyambali	_____	___/___/2026
CTO	Matome Mboweni	_____	___/___/2026

\1 1.0
\1 January 21, 2026
\1 Post-Phase 0 (March 2026)